



ENGINEERS LEVEL · SEMESTER 2

Smart Sorting AI

Student Workbook

Train an image classifier — and learn how models learn, and how bias creeps in.

Name: _____

THEORY

How Machines Learn

You train an image **model** by giving it **labelled examples (training data)**. It finds patterns and **classifies** new inputs with a **confidence** score. The ethics core: if the training data is unbalanced, the model shows **bias** — so what data you use, and whose data, matters. More balanced examples make fairer models.

Discuss: if a model only saw one kind of example, how might it be unfair — and how would you fix it?

Reason it

A model trained on 90 cats and 3 dogs is asked to classify a dog.

What is it likely to guess, and why? _____

Code we use:

model training data label classify confidence bias check

Words we are learning

- **Model** — a system that learns patterns from examples.
- **Training data** — the labelled examples a model learns from.
- **Label** — the category name attached to an example.
- **Classify** — the model's guess for a new input, with a **confidence**.
- **Bias** — unfair results caused by unbalanced or unrepresentative data.

Name: _____ Date: _____

Training Data & Labels

A model learns from **labelled** examples. Match and answer.

Label	the category name for an example
Training data	the examples the model learns from
More balanced examples	a fairer model

Why do more (and more varied) examples usually help?

Name: _____ Date: _____

Classify & Confidence

A model outputs a guess and a **confidence**. Interpret the results.

Cat 92%	confident it is a cat
Dog 51% / Cat 49%	unsure — nearly a coin flip
No examples like this	unreliable guess

If confidence is near 50/50, what should you do?

Name: _____ Date: _____

Spot The Bias

Unbalanced data causes **bias**. Identify the problem and a fix.

A recycling classifier trained mostly on plastic bottles keeps guessing 'plastic'.

What is the bias?

How would you fix the training data?

Name: _____ **Date:** _____

Plan My Classifier

Plan a fair classifier and run an ethics check.

My classes (labels):

Examples per class (aim balanced):

Ethics check — whose data, any unfairness?

Name: _____ **Date:** _____

PROJECT

Smart Sorting AI — Plan

Train an image classifier (e.g., sorting recyclables), test it on new images, and explain how it learns — then identify one bias risk and how balanced data reduces it.

1. What will my model sort?

2. Classes / labels:

3. How many examples per class?

4. One bias risk and my fix:
